HGFX supplementary evidence and provenance

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S1 Frozen identities and regeneration

The evaluated HGFX release is 1.0.0. Its source commit is

`4dd8fbd8239d05f2c7932a9a9b3b7795f0a9ab27`.

The MATLAB oracle is 8.2.0 at

`2437f4dc241541072722a2695ddeca7b44d83dd3`. These identities define the scope; current development features are not included. Original numerical experiments were not rerun for this revision.

Candidate figures regenerate with `python generate_figures.py` from this directory. The agreement/accuracy figure is redesigned from the frozen decision; recovery is redrawn in grayscale from the decision, and pyhgf includes a signed residual from all frozen trials. Horizon and backend figures are byte-identical copies of historical images. Their input/output hashes are recorded separately. The workflow diagram is conceptual and contains no data. Complete machine-readable source links appear in the repository claim ledger; this supplement supplies a compact reading route.

S2 Fitting and recovery

The official stress-fit D02 and holdout D08 are preserved failures, classified as matched reference limitations, not direct parameter parity. Their decision files and frozen source runs remain in the reference validation directory. No endpoint was forced, no trial was silently discarded and no tolerance was changed here. The historical M18 scientific failure remains separate from paired S7 accounting.

The paired grid has 72 parameter cases and 36 model-recovery datasets with three candidate fits each. There are 12 shards, two trial lengths (128,256), two truth scales (0.15,0.35 prior SD), six parameter replicates per stratum and three model replicates per stratum. The selection rule is minimum BIC. Metrics rounded for display; full precision remains in the frozen decision:

Model	Implementation	Convergence	Median r	Median sRMSE
hgf_binary	matlab	0.833333333333	0.202684570158	2.60966284547
hgf_binary	hgfx	0.833333333333	0.202684570141	2.60966284551
ehgf_binary	matlab	0.916666666667	0.461918369101	2.35912688823
ehgf_binary	hgfx	0.916666666667	0.461918369078	2.35912688867
uhgf_binary	matlab	0.791666666667	0.380968313583	2.95788661965
uhgf_binary	hgfx	0.791666666667	0.380968313591	2.95788661807

The frozen scientific criteria are convergence >=0.80, median r >=0.50, median sRMSE <=1.00 and model-recovery balanced accuracy >=0.50. Agreement is 36/36 winners, while balanced accuracy is 0.5833333333333334 in each implementation. These are deterministic grid summaries, not uncertainty intervals.

S3 Trial-horizon integrity

The original run 35272347167 contains 24 shards, 144 parameter cases and 72 model cases. The overall gate is false. Both winner agreement and the free-index contract fail. Ten BIC winners differ, confined to classic HGF at 512/1024 trials; the aggregate per-model mismatch classification covers all three families. Invalid classic-HGF trajectories remain in the denominator; undefined metrics remain gaps. A favorable diagnostic row is not an aggregate scientific success.

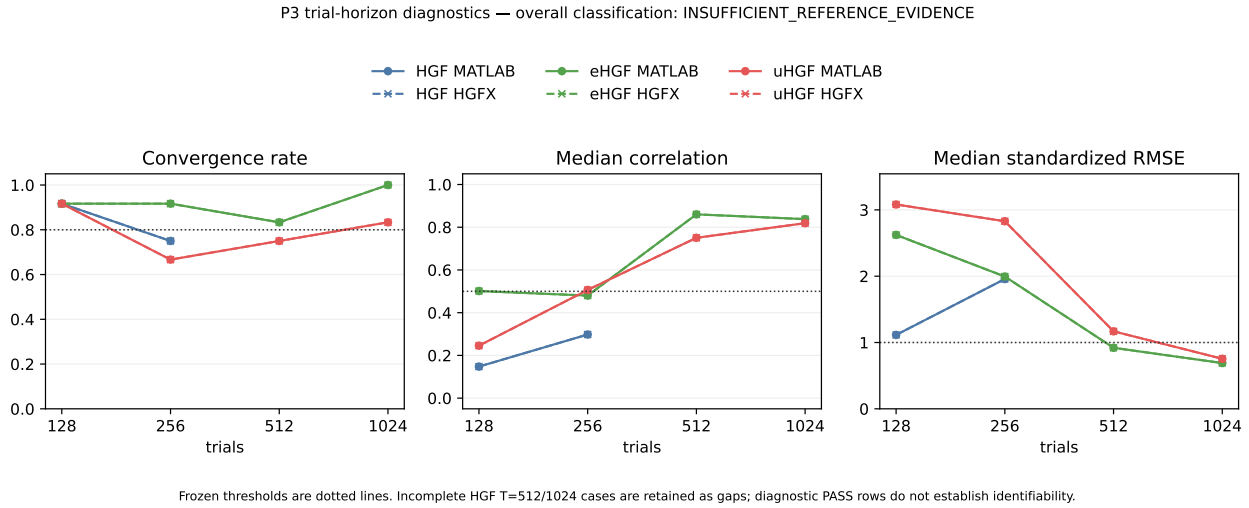


Fig. S1: Frozen diagnostic recovery metrics by horizon. Dotted lines are predeclared thresholds, gaps preserve invalid classic-HGF cases, and the paired integrity gate fails. No new uncertainty interval, interpolation or imputation is introduced

S4 Backend applicability

Physical evidence is retained from source `07b45a569e04e8e71244c5310dd2cc53dbb2b0ec`, Python 3.12.13, JAX/JAXLIB 0.11.1. Two Tesla T4 devices were visible; four required fitting cells ran on cuda:0. Residency checks and enumeration are separate. Its raw JSON and retention basis remain archived; it is not a new post-remediation GPU execution. The CPU remeasurement has its own source/runtime identity. The maximum gaps are 0.006783711260709424 (compatibility/JAX CPU, criterion 0.10) and 1.4210854715202004e-14 (JAX CPU/T4, criterion 1e-7).

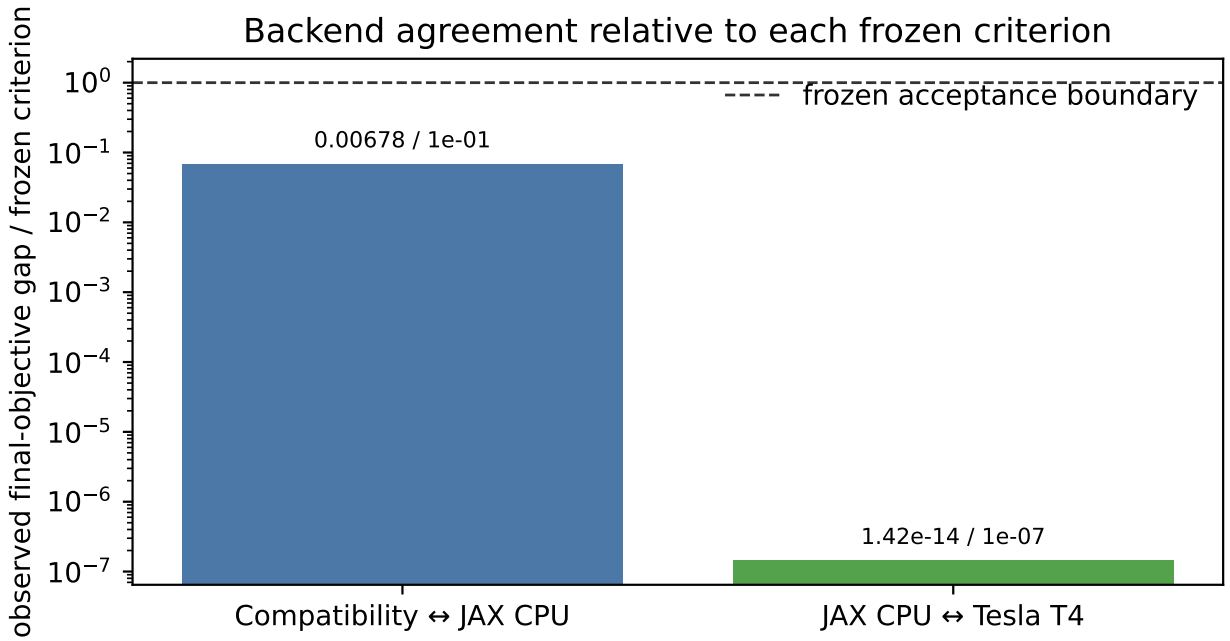


Fig. S2: Each maximum objective gap divided by its own frozen criterion; the dashed line is the acceptance boundary. These bars describe numerical agreement in recorded cells, without error bars. There is no speed measurement

S5 pyhgf common scope

pyhgf 0.3.2 source is `ccd43db5ee5abe4a5a35077d098e53cce2c070c2`. The fixed 128-trial case uses a three-level binary hierarchy, standard volatility updates and mean-field updates, zero drift and unit coupling. The native response inverse temperature is 48. Input and response hashes and the prospective semantic mapping remain in the original common-scope record. Field-wise measured errors:

Quantity	Classification	Maximum absolute error
derived first level input surprise	PASS	4.440892098500626e-16
derived first level prediction error	PASS	1.1102230246251565e-16
first level predicted probability	PASS	1.1102230246251565e-16
level 2 posterior mean	PASS	4.440892098500626e-16
level 2 posterior precision	PASS	6.661338147750939e-16
level 2 predicted mean	PASS	4.440892098500626e-16
level 2 predicted precision	PASS	6.661338147750939e-16
level 3 posterior mean	PASS	1.1102230246251565e-16
level 3 posterior precision	PASS	1.5543122344752192e-15
level 3 predicted mean	PASS	1.1102230246251565e-16
level 3 predicted precision	PASS	1.5543122344752192e-15
participant response nll per trial	NDC	not applicable
participant response nll total	NDC	not applicable

NDC applies to participant-response NLL at a probability boundary; belief agreement does not repair the response comparison. No clipping or changed tolerance was introduced after results.

Package-wide or fitted-model equivalence is untested.

S6 Applied feasibility and source routes

Pain source commit is `3f1d32aff73826a1385c359ca638ad0e11a7fda7`; all 504 tracked files are hashed in its source manifest. Its exact likelihood/config license and historical environment remain unresolved. Some simulation files carry GPL notices; no source was translated into MIT code. The missing plotting helper and filename/declaration mismatches remain recorded. Input-name confusion was resolved using the actual source table path, without manufacturing data.

SPIRL source commit is `86fd5122a744acfb5bc7ba12720426d8b4aa0ffa`. Its custom eHGF and seven joint choice/RT response models require a separate migration. The recorded original settings include 399 random starts and MATLAB R2019b; using standard eHGF or dropping RT is a changed analysis.

No published-study fit or new recovery was executed. The clean-wheel quickstart is synthetic workflow evidence only. Unlicensed external files remain local and ignored; public manifests contain hashes and URLs rather than participant values. The historical acquisition dossier remains unchanged.

S7 Evidence index and claims

The revision's CLAIM_EVIDENCE.csv maps manuscript paragraphs to source artifacts, hashes, locators and scope. Agent checks are distinct from human verification; author confirmation is pending throughout. The baseline hash inventory protects historical evidence. Paper protocol 1 remains the numerical authority. Failed recovery, sensitive fits, P3 integrity failure and response NDC remain in the main text. Editorial files record unresolved financial and authorship conditions.